



Muhammad Fahad Arshad

POSTDOCTORAL RESEARCHER · PHD, ENGINEERING THERMOPHYSICS

Ben Guerir, Morocco

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“Advancing sustainable energy solutions through computational catalysis and hydrogen production research”

Summary

Computational catalysis researcher blending heterogeneous experimentation with high-performance modelling (DFT, MD, AI/ML) to design low-carbon hydrogen and CO₂ utilisation pathways. Delivered reactor innovations, cross-functional collaborations, and data-driven workflows that accelerated catalyst discovery from concept to pilot scale.

Education

New Technology Lab, University of Chinese Academy of Sciences

Beijing, China

PHD, ENGINEERING THERMOPHYSICS

Aug 2017–Aug 2023

- Thesis: *Investigation on catalysis and surface thermochemistry of La-based transition metal oxide catalysts*
- Built coupled experimental–computational workflows for heterogeneous and photo-catalytic systems, spanning DFT, micro-kinetics, and lab-scale reactor design.
- Created bespoke photocatalytic and photolysis reactors that increased quantum efficiency by up to 35% versus legacy rigs.
- Authored 8+ peer-reviewed articles through collaborations with CAS institutes and international partners; dissertation centred on data-driven catalyst discovery for sustainable energy.

University of Engineering and Technology

Lahore, Pakistan

BSC, MECHANICAL ENGINEERING (FLUID DYNAMICS)

Aug 2012–Aug 2016

- Graduated with specialisation in fluid dynamics; capstone coursework in thermodynamics, heat transfer, and energy systems.
- Completed industry-sponsored design projects utilising CAD/CAM, CFD, and experimental validation of thermal processes.

Professional Experience

University Mohammed VI Polytechnic (UM6P)

Ben Guerir, Morocco

POSTDOCTORAL RESEARCHER, CHEMICAL AND BIOCHEMICAL SCIENCES

2025–Apr 2026

- Steer multi-institutional hydrogen production and CO₂ valorisation programmes from concept to pilot scale, aligning project milestones with national energy-transition initiatives.
- Build automated DFT, molecular dynamics, and micro-kinetic pipelines on UM6P’s HPC cluster to interrogate catalyst stability and reaction pathways.
- Deploy machine learning and LLM tools to shortlist catalyst formulations, reducing experimental screening time by ~40%.
- Develop agentic AI frameworks (OpenContextFramework) and scientific data crawlers for automated literature mining and catalyst discovery.
- Conduct high-pressure autoclave experiments with biomass feedstocks; prepare and characterize zeolite-based catalysts for sustainable fuel production.

Institute of Engineering Thermophysics, Chinese Academy of Sciences

Beijing, China

POSTDOCTORAL RESEARCHER

2023–2024

- Mentored a cohort of 5+ international researchers, instituting lab protocols that improved catalytic test repeatability by 25%.
- Co-led CO oxidation and methane activation campaigns, delivering three peer-reviewed papers and two invited talks.
- Managed partnerships and instrumentation sharing across CAS institutes, doubling collaborative project inflow.

Pakistan Industrial Technical Assistance Centre (PITAC)

Lahore, Pakistan

INTERN ENGINEER

2016–2017

- Supported CAD/CAM workflows and produced fabrication drawings for bespoke tooling projects.
- Documented test procedures and liaised with senior engineers to implement design refinements for thermal systems.

Skills

Computational	DFT (Quantum ESPRESSO, VASP, Gaussian); MD (LAMMPS); Microkinetic Modeling (Cantera, Chemkin); Python (NumPy, Pandas, scikit-learn, ASE, pymatgen); Rust; HPC; Git, Bash, 终端
Experimental/Analytical	XRD, XPS, Raman, BET, SEM/EDS; Sol-gel, CVD; Zeolite synthesis; High-pressure autoclave experiments; Biomass conversion; Oxidation, Photocatalysis
AI/ML & LLM Development	Machine Learning for catalysis; LLM tool development; Scientific data crawlers/downloaders; Agentic frameworks (OpenContextFramework)
Research Interests	Hydrogen production, CO ₂ valorization, Heterogeneous catalysis, Photocatalysis, Zeolite catalysts, Biomass conversion, ML-accelerated catalyst discovery
Software Development	Autoclave pressure calculator; oa_corpus downloader; scrysrust (Rust); OpenContextFramework (agentic AI)
General/Technical	Technical writing, Adobe Illustrator vector drawing, CAD/CAM, Application studies, Research supervision
Languages	English (Fluent), Urdu (Native), Mandarin (Conversational)

Software Development

- **Autoclave Pressure Calculator** (Personal Project) – Engineering tool for calculating pressure parameters in high-pressure autoclave experiments for safe operation with biomass and catalytic systems.
- **oa_corpus** (Personal Project) – Scientific publication downloader and corpus builder for automated literature mining and large-scale text analysis.
- **scrysrust** (Personal Project) – Rust-based molecular modelling software for computational chemistry; leverages Rust’s performance for DFT pre/post-processing.
- **OpenContextFramework** (Personal Project) – Agentic AI framework for scientific workflows; enables automated catalyst screening and literature-informed hypothesis generation.

Selected Publications

Citation Metrics: 178 citations | h-index: 7 | i10-index: 6 (Google Scholar)

Selected Publications (11 total)

First-author publications marked with *

- * Yousuf M, **Arshad MF**, Tian Z-Y. (2024). Thermodynamic properties calculations of Cu-based species. *Int J Chem Kinet.* 56: 310–322.
- El Kasmi A, **Arshad MF**, Waqas M, *et al.* (2023). Insights into catalytic oxidation of CO over Cu catalyst. *Materials Research Bulletin*, 112343.
- * **Arshad MF**, El Kasmi A, Fonzeu Monguen CK, *et al.* (2023). Insights into the Role of Carbonates and Hydroxides in La–Co Oxides for CO Catalytic Conversion. *ES Energy & Environment* 20: 908.
- Fonzeu Monguen CK, El Kasmi A, **Arshad MF**, *et al.* (2022). Oxidative Dehydrogenation of Propane into Propene over Chromium Oxides. *Ind & Eng Chem Research*, 61(13), 4546–4560.
- * **Arshad MF**, Wu LN, El Kasmi A, *et al.* (2021). Ab Initio Calculation of Surface Thermochemistry for Popular Solid Transition Metal–Based Species. *ACS Omega*, 6(35), 22525–22536.
- * **Arshad MF**, El Kasmi A, Waqas M, Tian ZY. (2021). Insight into one–step synthesis of active amorphous La–Co thin films for catalytic oxidation of CO. *Applications in Energy and Combustion Science*, 5, 100021.
- Wu LN, Tian ZY, Kasmi A el, **Arshad MF**, Qin W. (2021). Mechanistic study of the CO oxidation reaction on the CuO (111) surface. *Proceedings of the Combustion Institute*, 38(4), 5289–5297.
- Waqas M, El Kasmi A, Wu LN, **Arshad MF**, *et al.* (2021). Catalytic combustion of CO over Cu–doped iron oxides: CO₂ effects on activity. *Fuel*, 289, 119760.

Full publication list available on:

- **Google Scholar:** scholar.google.com/citations?user=fwiNZasAAAAJ
- **ORCID:** orcid.org/0000-0003-1828-9458

Conferences & Presentations

European Combustion Meeting

Australia

AB INITIO CALCULATION OF SURFACE THERMOCHEMISTRY FOR POPULAR SOLID TRANSITION METAL–BASED SPECIES

2020

- Authors: Arshad MF, Wu LN, El Kasmi A, Qin W, Tian ZY
- **Oral Presentation**

Chinese Combustion Symposium

Tianjin, China

INSIGHT INTO ONE–STEP SYNTHESIS OF ACTIVE AMORPHOUS LA–CO THIN FILMS FOR CATALYTIC OXIDATION OF CO

2020

- Authors: Arshad MF, El Kasmi A, Waqas M, Tian ZY
- **Oral Presentation**

Teaching Experience

University of Chinese Academy of Sciences

Beijing, China

GRADUATE TEACHING ASSISTANT

2020–2021

- Taught postgraduate students and assisted with course delivery, examinations, and grading.
- Provided mentoring to graduate students on research methodologies and supervised laboratory work.

University Mohammed VI Polytechnic (UM6P)

Ben Guerir, Morocco

LECTURER - PROCESS ENGINEERING

2025–2026

- Delivered lectures on Process Engineering to undergraduate/graduate students.
- Covered topics including reactor design, process optimization, and safety in chemical processes.

Awards & Honors

2022 **Outstanding International Student**, UCAS Beijing, China

Beijing, China

2020–2024 **CAS–TWAS Fellowship**, The World Academy of Sciences

Beijing, China

2017–2020 **Belt and Road Fellowship**, UCAS

Beijing, China

2016 **Top Crisis Manager**, University of Engineering and Technology

Lahore, Pakistan

Leadership & Services

Institute of Engineering Thermophysics, Chinese Academy of Sciences

Beijing, China

LEADER OF INTERNATIONAL STAFF AND STUDENTS

2021–2024

- Organized weekly meetings to coordinate research activities and foster collaboration among international researchers.
- Provided comprehensive support for research projects, experiments, and computational calculations.
- Facilitated knowledge transfer and skills development within the international research community.
- Served as liaison between international researchers and laboratory management.

University of Chinese Academy of Sciences – International Student Office

Beijing, China

STUDENT GUIDE AND TRANSLATOR

2017–2023

- Guided international students and assisted in organizing orientation programs and cultural events.
- Provided translation services from Mandarin to English for academic and administrative purposes.
- Helped international students navigate university systems and integrate into academic life.
- Contributed to improving cross-cultural communication and understanding.

Digital Identifier Links & Metrics

Research Profiles:

- **ORCID:** 0000–0003–1828–9458
- **Scopus ID:** 57219161795
- **Google Scholar:** fwINZasAAAAJ
- **Web of Science Researcher ID:** rid102402
- **ResearchGate:** Muhammad–Fahad–Arshad

Google Scholar Metrics:

- **Citations:** 178 (all time)
- **h-index:** 7
- **i10-index:** 6

References: Available upon request. Please contact for details.